Fiscal Year:	FY 2004	**Task Last Updated:**	FY 10/07/2004
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Project Title:	Senseless and Gfi1 in sensory neuron and hair cell development		
Division Name:	Space Biology		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Human Research Program Risks:	None		
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Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
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Task Description:	Hair cells function as mechanoreceptors to transduce sound and gravity. We have previously shown that the generation of the inner hair cells in the balance organs and the organ of Corti are dependent on the expression of the Mouse atonal homologue 1 (Math1). In fruit flies, Atonal is required for specification of mechanoreceptors involved in hearing and gravity sensing. We recently isolated a novel gene in Drosophila, senseless, whose expression is dependent on the proper expression of the proneural bHLH genes, including atonal. Senseless in turn is required for the upregulation and maintenance of proneural gene expression in the sensory organ precursors (SOP) and for proper sensory organ development. In addition, we have shown that senseless synergizes with the proneural genes, as co-expression of low levels of Atonal and Senseless leads to massive recruitment of additional SOPs and sensory organs from ectodermal cells. Hence, Drosophila senseless is both necessary and sufficient for sensory organ development. The ability of Senseless to strongly potentiate the effects of Atonal in fruit flies implies that co-expression of a vertebrate Senseless homologue in combination with Math1 may greatly facilitate hair cell recruitment in vertebrates as well. Based on sequence homology, mammals contain two senseless homologues: Gfi1 and Gfi1b. We present evidence that Gfi1 expression is mostly restricted to precursors of the PNS as it is expressed in the dorsal root ganglia, the PNS ganglia of the head, and the developing epithelia of the ear. Here, we propose to identify new genes that play a role in the specification of mechanoreceptors in Drosophila and to study the potential conservation of this pathway in mice. We believe that these data will provide important clues about the molecular mechanisms of hair cell specification and that these tools may be exploited to (re)generate hair cells in animals in the near future.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is important for two main reasons. First, it will help us understand how hair cells are formed, i.e. what are the earliest molecular mechanisms that instruct an epithelial cell to become a hair cell. Second, this basic knowledge may allow us to regenerate hair cells in inner ears of animals that have lost the hair cells of the balance and gravity sensing organs. The proof of principle has now been established as in the last year, two teams have shown that Math1 expression is able to regenerate hair cells, and that these hair cells attract growth cones of nerves to be re-innervated (Kawamato et al., 2003; Shou et al., 2003). We believe that coexpression of Math1 and Gfi-1 may recruit hair cells much more efficiently based on our observations in fruit flies.
Task Progress:	In Drosophila we are currently studying three different aspects of senseless biology. In the first project, we have defined the regulatory elements that control the precise expression pattern of senseless. These regulatory elements have been dissected into relatively small pieces that contain between 300 and 400 nucleotides. They recapitulate almost the entire expression pattern of senseless in the embryo and in imaginal discs. The relevant DNA sequences contain binding sites for basic helix-loop-helix (bHLH) proteins of the achaete-scute family and enhancer of split (E(spl)) proteins. Mutational analysis shows that these sequence motifs are required for the binding of the proneural proteins to the senseless enhancer in vitro. In addition, point mutations in these conserved sequences (E boxes) strongly reduce expression of the senseless-lacZ marker in vivo, suggesting that the proneural binding sites are required in vivo for the proper expression of the senseless gene. We therefore conclude that the proneural genes like atonal, achaete and scute play a role in the direct regulation of senseless transcription (Jafar-Nejad et al., 2003). We have also shown that senseless and E(spl) (Enhancer of split proteins) display an antagonistic relationship: high levels of senseless suppress E(spl) function and vice-versa. This antagonistic relationship is in part mediated by transcription, in part by protein-protein interactions. Our data indicate that E(spl) can transcriptionally suppress expression of sens-lacZ. In addition, we have shown that a 25 aa domain in Senseless is able to bind helix 3 and helix 4 of E(spl) proteins. We believe that this protein-protein interaction is antagonistic to the function of both proteins (Jafar-Nejad et al., 2003). To determine how senseless interacts with the proneural proteins we have developed a transcription assay in S2 cells. Our data show that senseless binds to a binding site in the achaete enhancer. This binding site is required for an inhibition of transcription of the daughterless-achaete synergism at low levels of transfection of senseless in S2 cells. At high levels of transfection of senseless, senseless shows a dramatic synergism with the daughterless-achaete gene products. We propose that low levels of senseless inhibit proneural gene expression in the cells surrounding the SOP, but that senseless strongly synergizes with proneural gene expression in the SOP. These data are supported by in vivo observation (Jafar-Nejad et al., 2003). We have also recently shown that dCTBP (Drosophila carboxy terminal binding protein) may play a pivotal role in this repression. The dCTBP protein is able to bind senseless, and overexpression of dCTBP causes loss of bristles in the notum. We are currently investigating the significance of these protein interactions. In mice, we have shown that the senseless homologue corresponds to Gfi1. Interestingly, human small cell lung cancers are derived from pulmonary cells with a neuroendocrine phenotype. They are driven to proliferate by autocrine and paracrine neuropeptide growth factor stimulation. The molecular basis of the neuroendocrine phenotype of lung carcinomas is relatively unknown. The Achaete-Scute Homolog-1 (Ash1) transcription factor is critically required for the formation of pulmonary neuroendocrine cells and is a marker for human small cell lung cancers. The Drosophila orthologs of Ash1 (Achaete and Scute) and the Growth factor independence-1 (Gfi1) oncoprotein (Senseless) genetically interact to inhibit Notch signaling and specify fly sensory organ development. In collaboration with Lee Grimes (University of Louisville) we have shown that Gfi1, like Ash1, is expressed in neuroendocrine lung cancer cell lines, and that Gfi1 in lung cancer cell lines functions as a DNA-binding transcriptional repressor protein. Forced expression of Gfi1 potentiates tumor formation of small lung cancer cells. In primary human lung cancer specimens, Gfi1 expression strongly correlates with expression of Ash1, the neuroendocrine growth factor gastrin-releasing peptide, and neuroendocrine markers synaptophysin and chromogranin A ($P < 0.0000001$). Gfi1 co-localizes with chromogranin A and calcitonin-gene-related peptide in embryonic and adult murine pulmonary neuroendocrine cells. In addition, mice with a mutation in Gfi1 display abnormal development of pulmonary neuroendocrine cells, indicating that Gfi1 is important for neuroendocrine differentiation (Wallis et al., 2002; Kazanian et al., 2004, in press). These and other projects on senseless and Gfi1 will be continued during the new three-year funding period. Indeed, NASA decided to continue funding for a new project which is the continuation of this grant. I am really grateful for your continued support!
Bibliography Type:	Description: (Last Updated: 09/03/2009)

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